Requirements Engineering (III)

Training for the IREB CPRE
Foundation Level test

Prof. Camille Salinesi

1h1/2

EU 3 – REQUIREMENTS ELICITATION



At the end of this chapter, you should

1.1 Know various types of requirements sources

1.2 Know the significance of requirements sources and the consequences of disregarded requirements sources

1.3 Know the most important information of the stakeholder documentation

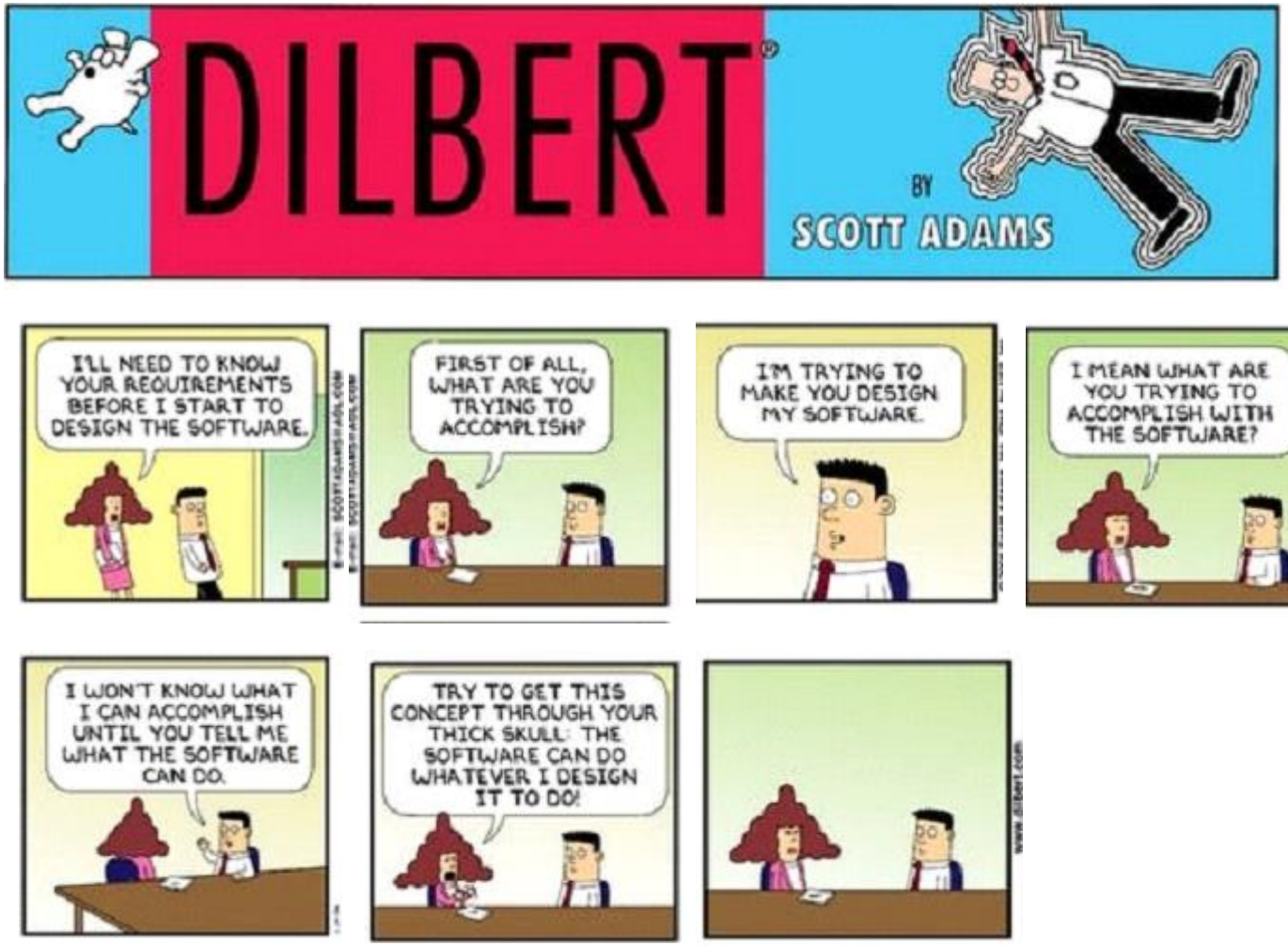
1.4 Know important principles in dealing with stakeholders (stakeholder rights and duties)

2.1 Master and using the content and significance of the Kano model

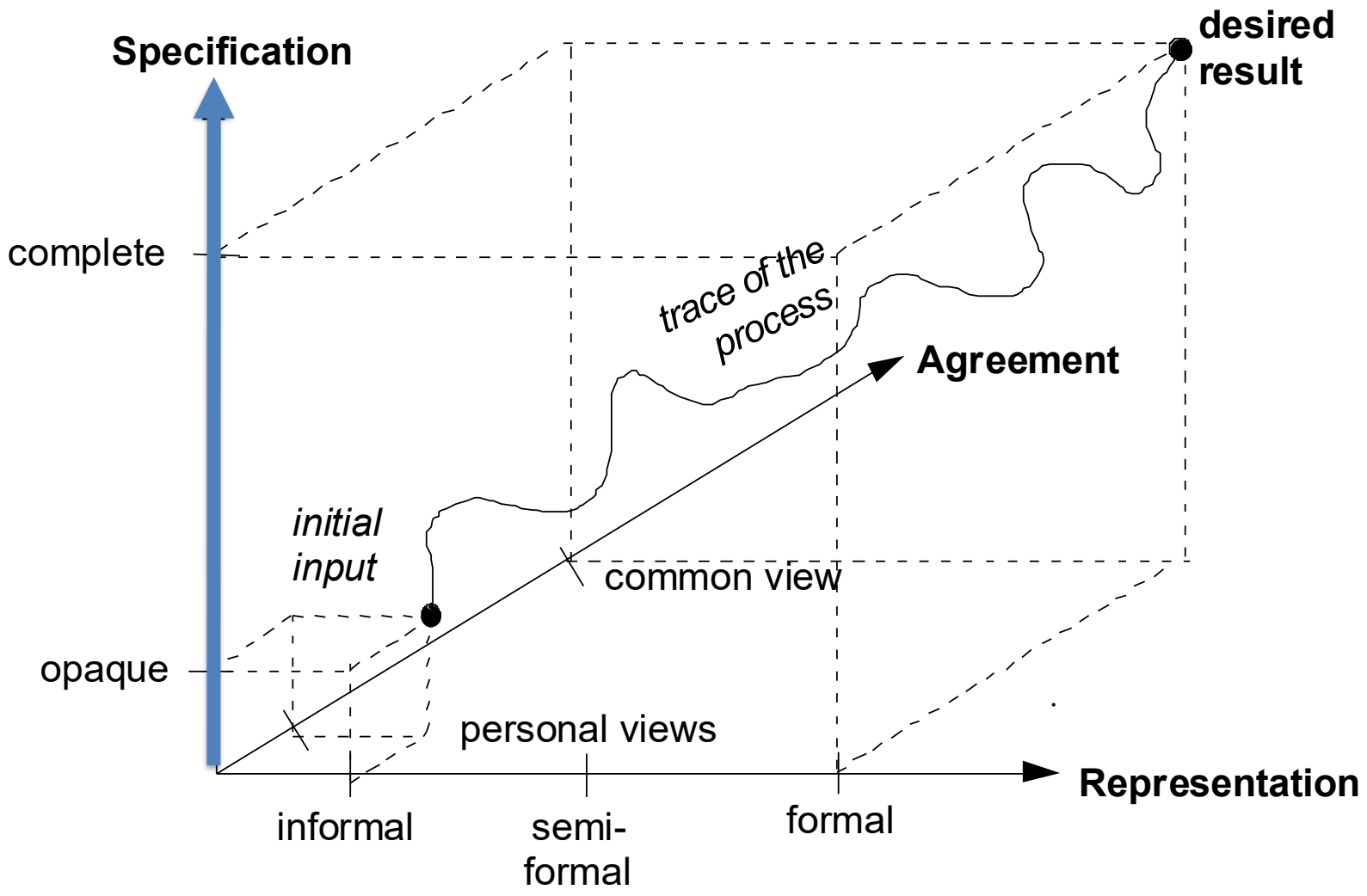
3.1 Know influencing factors for the choice of elicitation techniques

3.2 Know advantages and disadvantages of elicitation techniques

3.3 Master and use the following types of elicitation techniques and examples for each: survey techniques, creativity techniques, document-centered techniques, observation techniques and supporting techniques


DILBERT
BY
SCOTT ADAMS
I'LL NEED TO KNOW YOUR REQUIREMENTS BEFORE I START TO DESIGN THE SOFTWARE.
FIRST OF ALL, WHAT ARE YOU TRYING TO ACCOMPLISH?
I'M TRYING TO MAKE YOU DESIGN MY SOFTWARE.
I MEAN WHAT ARE YOU TRYING TO ACCOMPLISH WITH THE SOFTWARE?
I WON'T KNOW WHAT I CAN ACCOMPLISH UNTIL YOU TELL ME WHAT THE SOFTWARE CAN DO.
TRY TO GET THIS CONCEPT THROUGH YOUR THICK SKULL: THE SOFTWARE CAN DO WHATEVER I DESIGN IT TO DO!

The 3 dimensions of RE



A requirement statement is complete when…

- it "… describes the necessary capability, characteristic, constraint, or quality factor … without needing other information to understand the requirement"
  (INCOSE guide for writing requirements)

How do you get these goddam requirements ?



People

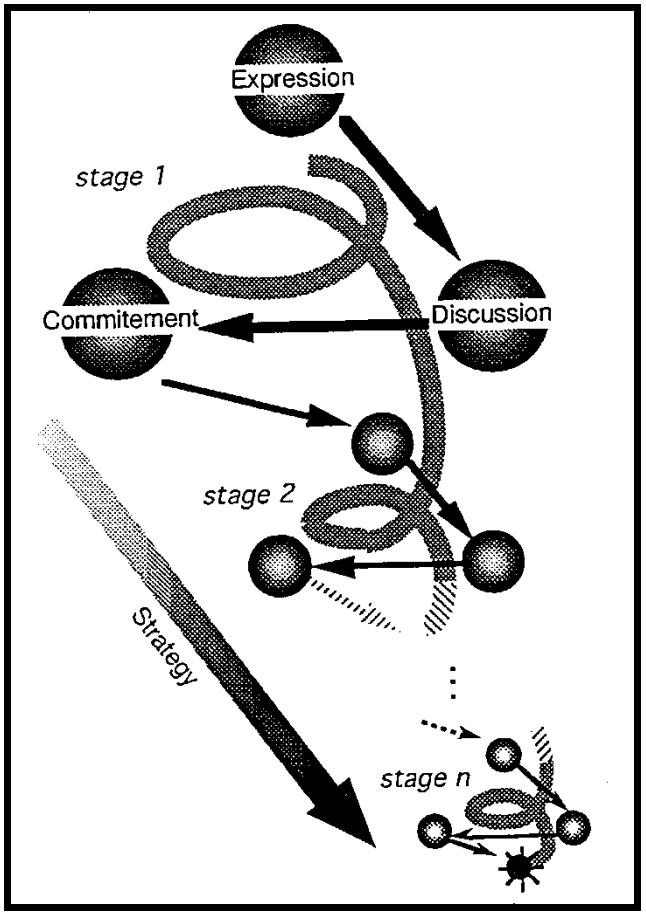
- don’t know what they want
- want a lot of things (Santa syndrom)
- mix up wishes, needs and requirements, problem-solution, their responsibility and the system’s responsibility, the system and its environment, etc.
- have difficulty expressing a need in a simple and understable way
- expect things that are not feasible
- etc.

Requirements are **not given**



- Remember the As-Is / To-Be framework
- The starting point for designing the To-Be is As-Is + change
- Context knowledge is a source
- Problems and goals are key

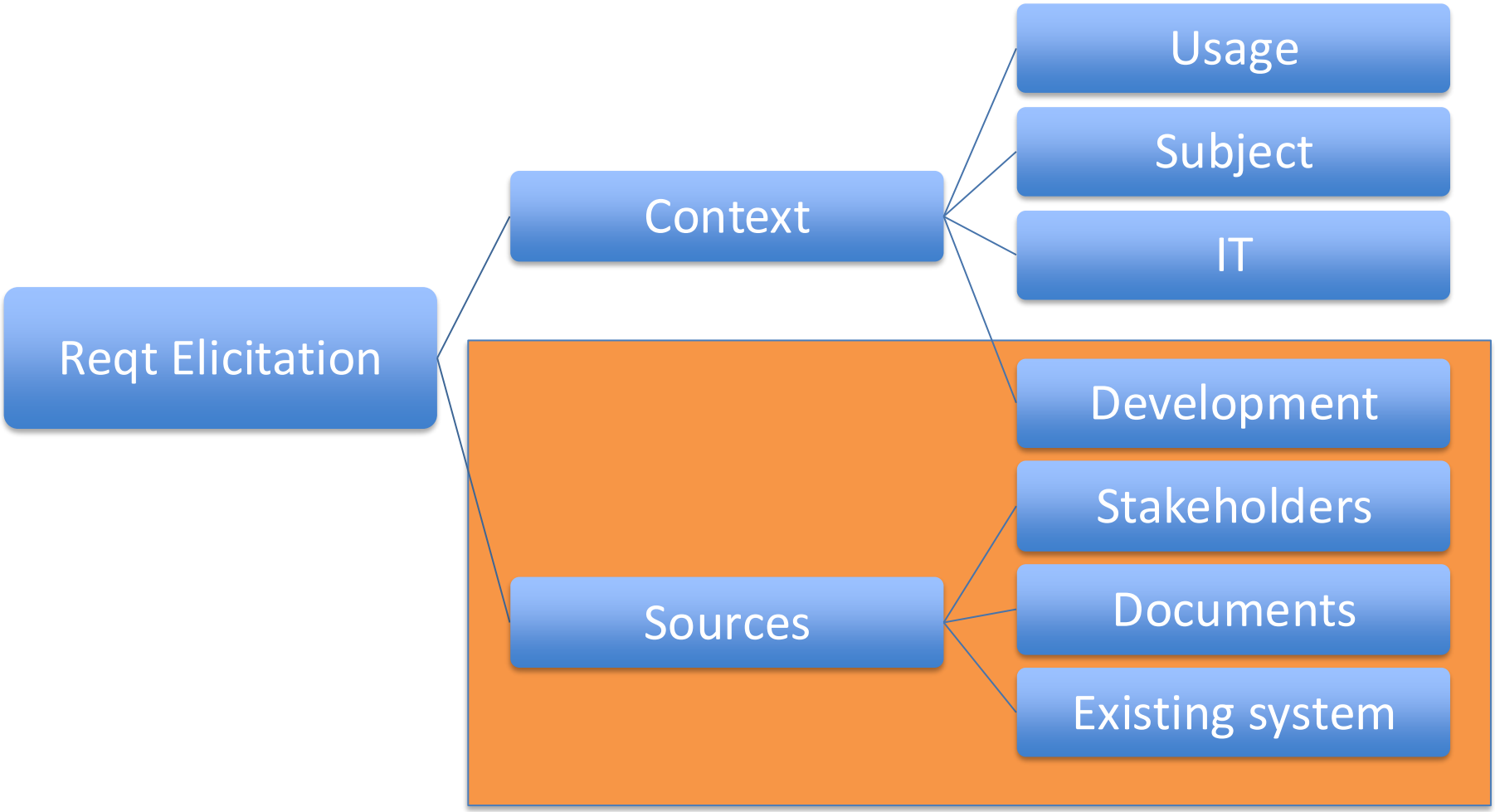
The “enquiry cycle”



Source : A. Anton, C Potts

- Expression
  - Common problems
  - Environment/context
  - Strategy
  - Goals
  - Scenarios
- Discussion
  - Interviews
  - Surveys
  - Online discussion
- Engagement
  - Change request
  - Negotiations
  - Priorisation

Foundations



UE 3

Requirements Elicittion

3.1 – DEFINE THE VISION

Start with the **end in mind**

- **Why**:
  - To define the target&scope
  - To coordinate and lead teams
  - To develop a shared understanding
- **What**:
  - Goals
  - The « why »
  - Strategic stakes
- **How**:
  - « Product vision»
  - « Business case »
  - « Mission »
  - Short and visual !

Make a **“Vision statement”**

« We choose to go to the Moon.

We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard, because that goal will serve to organize and measure the best of our energies and skills, because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one which we intend to win, and the others, too. »



J.F Kennedy

Septembre 12 1962

Rice University, Houston-Texas

Develop a « **Product vision** »



Apple iPad:

« best solution » to navigate on the web, read emails, wath pictures and videos »

Intelligent design, light and thin

Application marketplace

source: http://store.apple.com/

Elements of a Product Vision



1. **Where are we heading?**
   Problems, goals, markets, beneficiaries

2. **How do we go there?**
   Main functions, features, constraints

3. **Why will it succeed?**
   Milestones, risks, approach

Source [McGrah 2000] « Product Strategies for High-Tech Companies »

Example : **www.livre.fr**

1. **Where are we heading?**
   First online french speaking library

2. **How do we go there?**
   Better proximity with the market owing to synergy with 45% of French libraries and the leader online sales sites

3. **Why will it succeed?**
   Our team has a critical expertise on sales techniques over the net and social media marketing

Outcome of the Product Vision

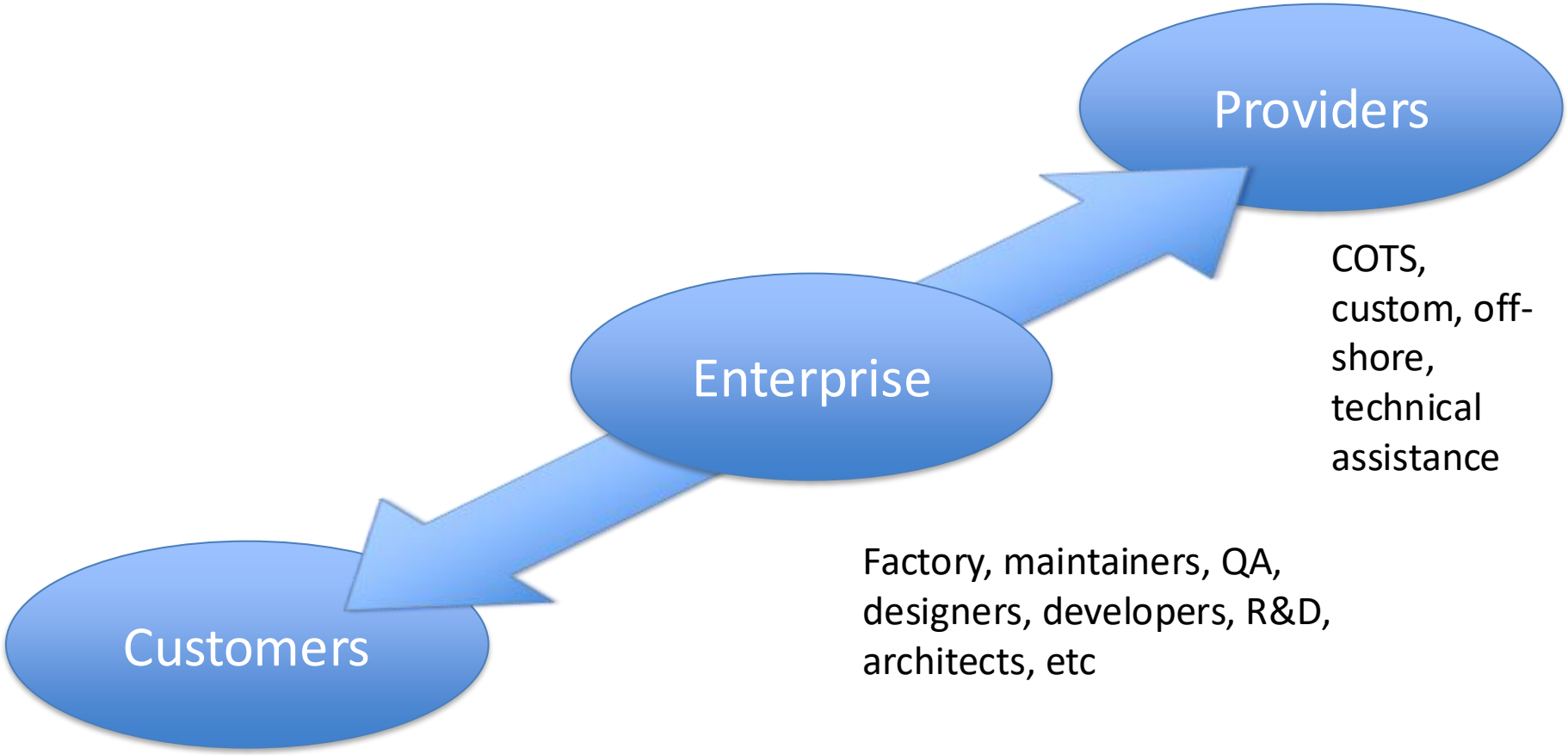
- Accelerate schedules
- Better goals lead to
  - Ask **more adequate questions** to the customers
  - Make sure providers and subcontractors have a **shared vision**
  - Improve **collaboration** among teams

UE 3

Requirements elicitation

3.2 – WORK WITH STAKEHOLDERS

Stakeholders



Personas

- *Class of users with an image, features, behaviour, etc*
- *Role playing*
- *Make types of users more concrete*
- *Provide diversified viewpoints on the system*
- *Also used in marketing and Extreme Programming*







Stakeholders cheklist



- Name
- Role
- Contact
- Availability
- Criticality for the project
- Domain and level of expertise
- Personal motivations/stakes

Animating stakeholders

Techniques

- Information meetings
- Interviews
- Brainstorming
- Focus groups
- Surveys

Attitudes

- Anticipate
- Motivate
- Collaborrate
- Respect roles



Customers' **rights**

As a customer, you can expect that

- Analysts speak your language
- Learn your domain, your goals
- Document your requirements and formalize them
- Explanations for important decision will be shared with you
- You will be involved in the production of ideas, dome decisions
- Have an opportunity to share your priorities, concerns,

[Wiegers 2003] « Software Requirements »

Customers' **responbilities**

As a customer you have the responsibility to :

- Train analysts in your domain
- Clarify terminology
- Be precise, explicit on your demands, priorities, concerns
- Respect others demands, priorities, concerns
- Provide insightful feedback when asked, eg. By reviewing documents, participating to evaluations, etc
- Comply with the project management rules and organisation

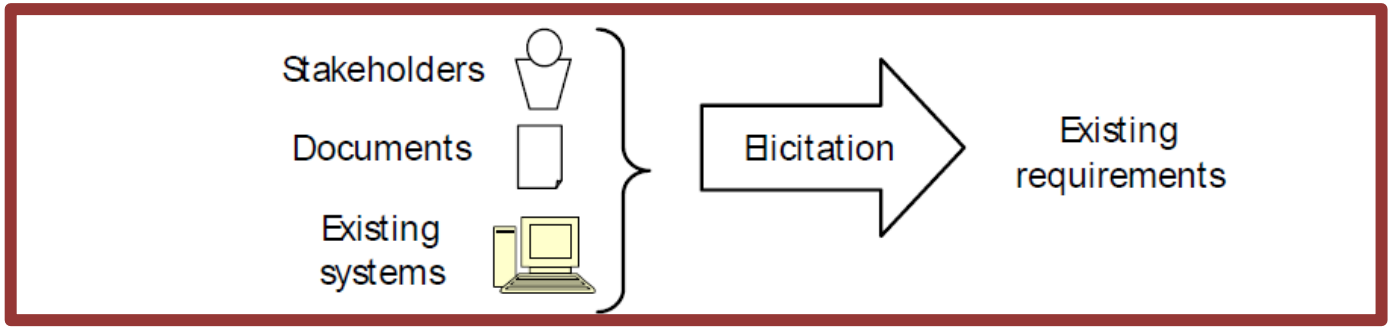
[Wiegers 2003] « Software Requirements »

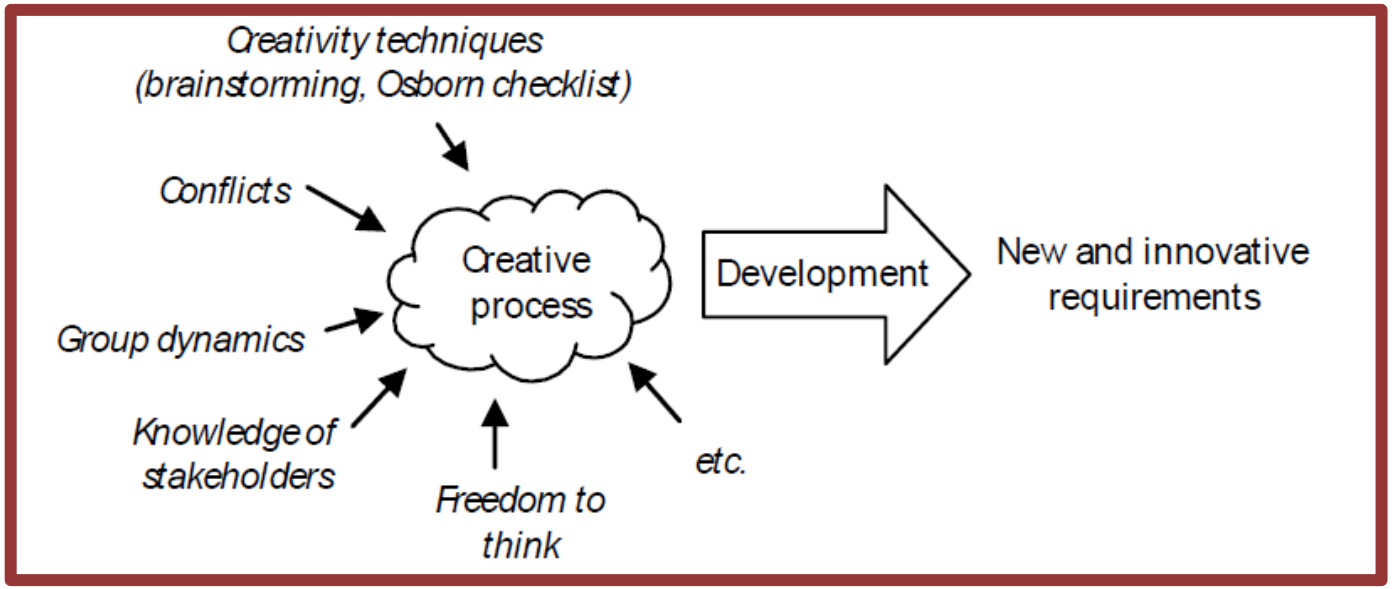
UE 3

Requirements Elicitation

3.3 – REQUIREMENTS ELICITATION TECHNIQUES

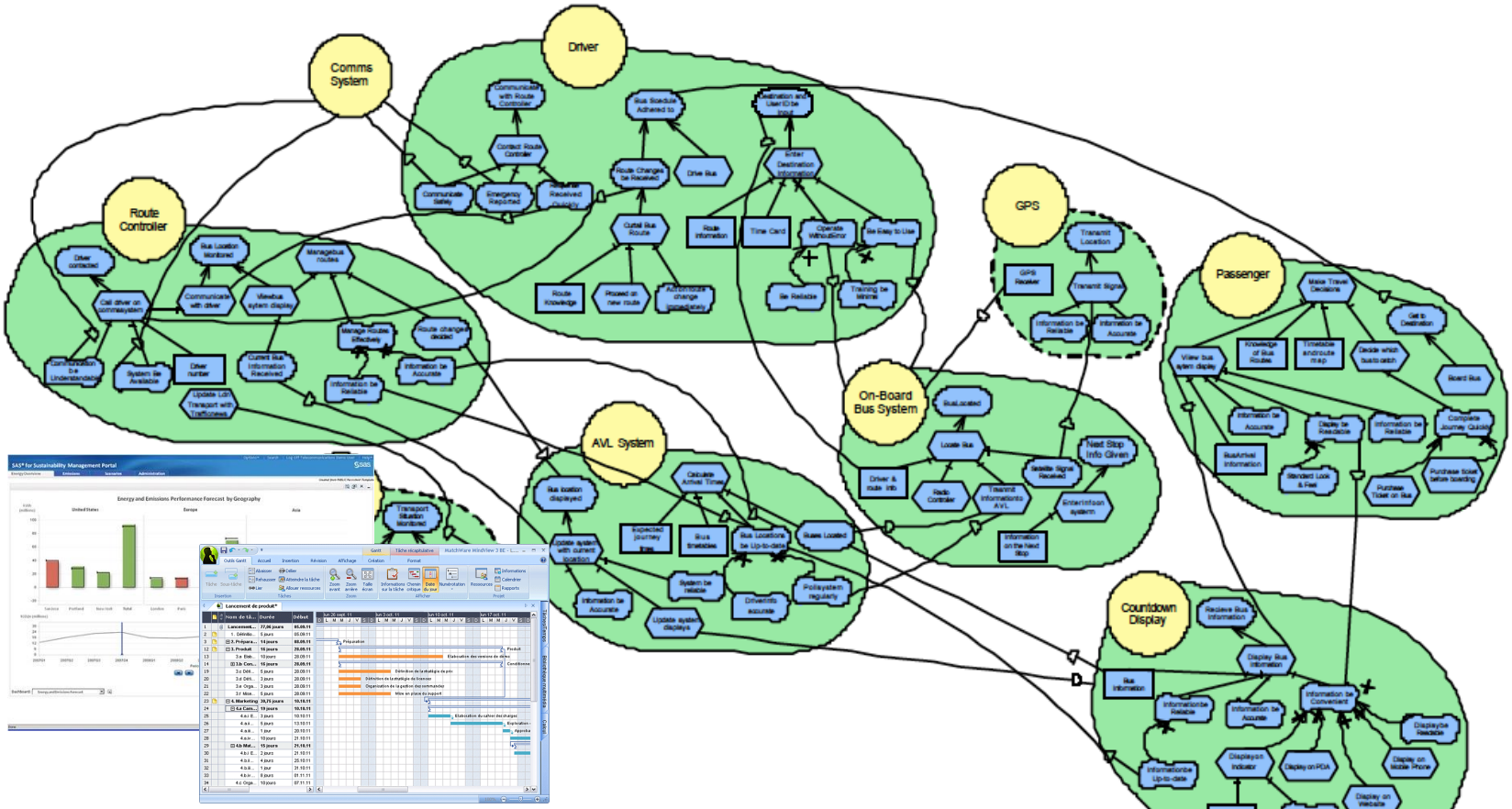
Elicitation techniques for **New VS Existing** requirements





[Pohl 2010] « Requirements Engineering » §21

« Early RE »
Setting up strategic goals



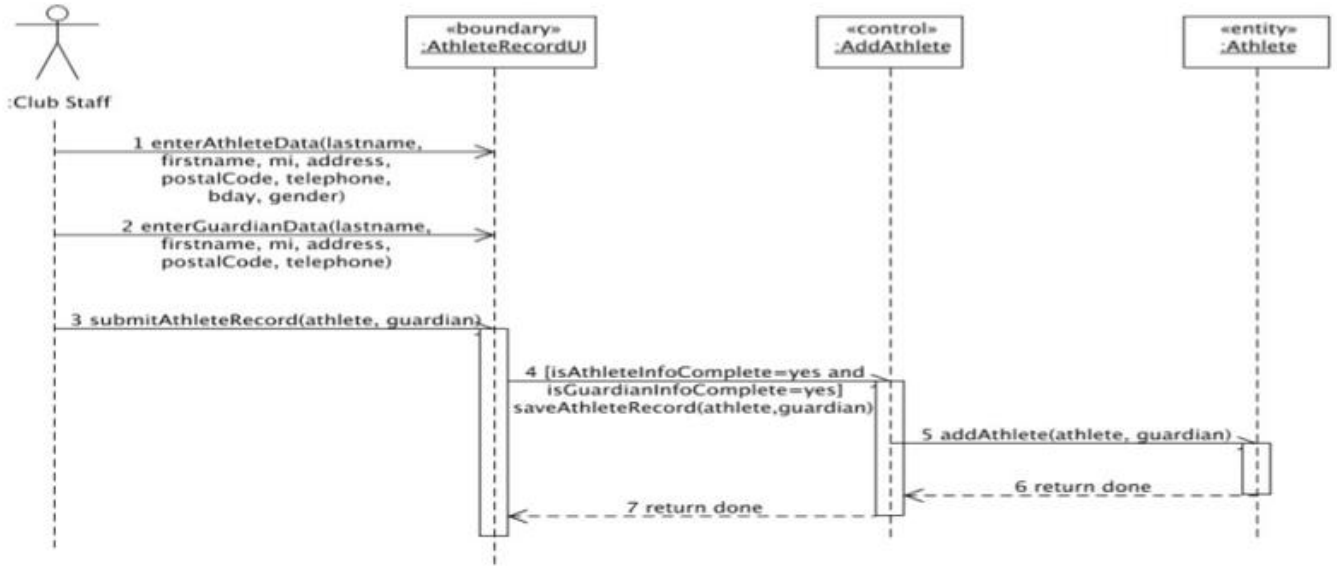
De S. Jones, N Maiden. « RESCUE: An Integrated Method for Specifying Requirements for Complex Socio- Technical Systems ».

Goal-based and **Scenario-**based requirements elicitation techniques

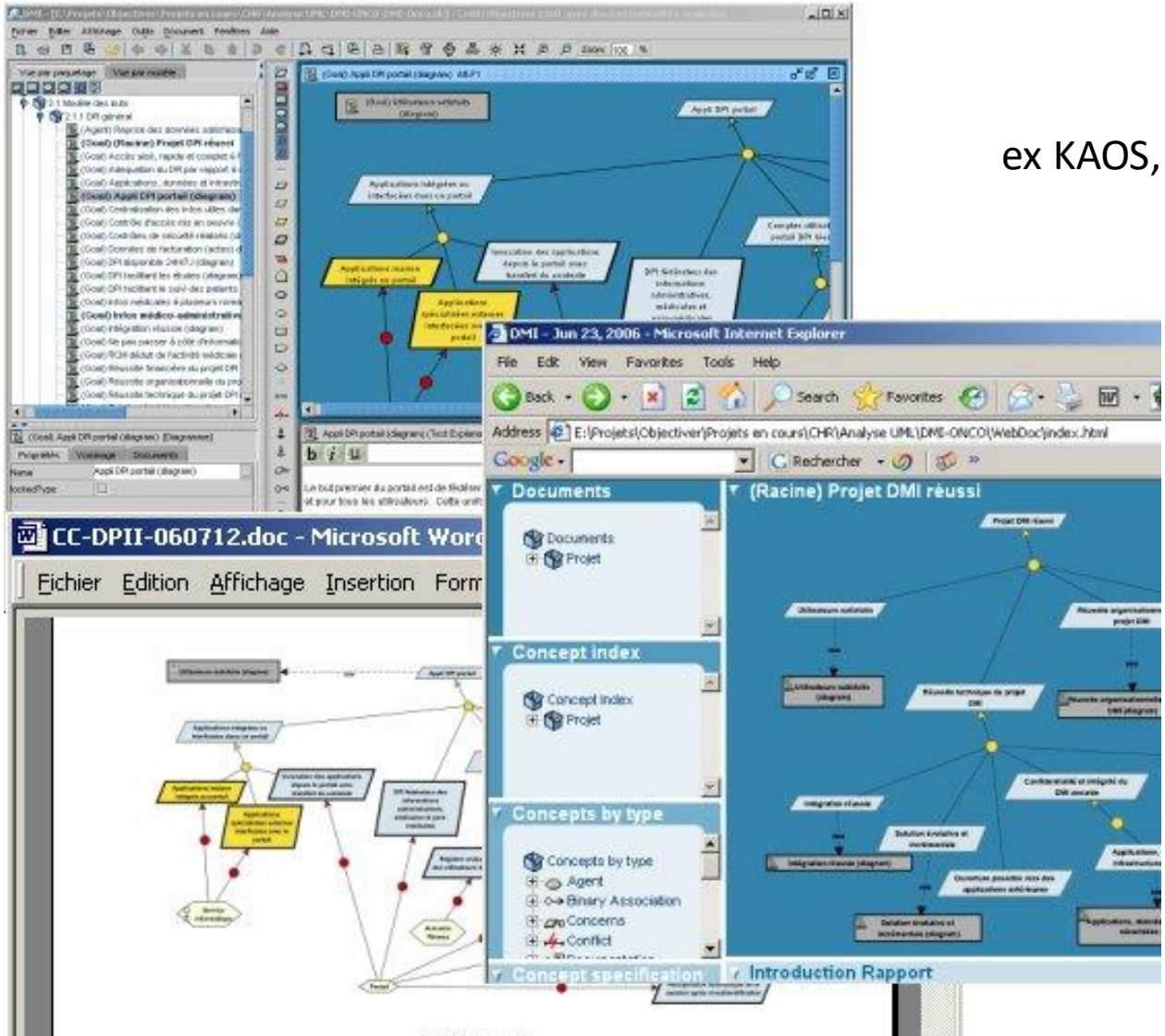
- Define goals
- Refine goals
- Contextualize with scenarios
- Develop innovating requirements from goals and scenarios

[Pohl 2010]
« Requirements Engineering » §21.3

Scenarios


:Club Staff
«boundary»
:AthleteRecordUI
«control»
:AddAthlete
«entity»
:Athlete
1 enterAthleteData(lastname, firstname, mi, address, postalCode, telephone, bday, gender)
2 enterGuardianData(lastname, firstname, mi, address, postalCode, telephone)
3 submitAthleteRecord(athlete, guardian)
4 [isAthleteInfoComplete=yes and isGuardianInfoComplete=yes] saveAthleteRecord(athlete,guardian)
5 addAthlete(athlete, guardian)
6 return done
7 return done

AND/OR goal refinement



ex KAOS, Objectiver

Bottom-up / Top-down

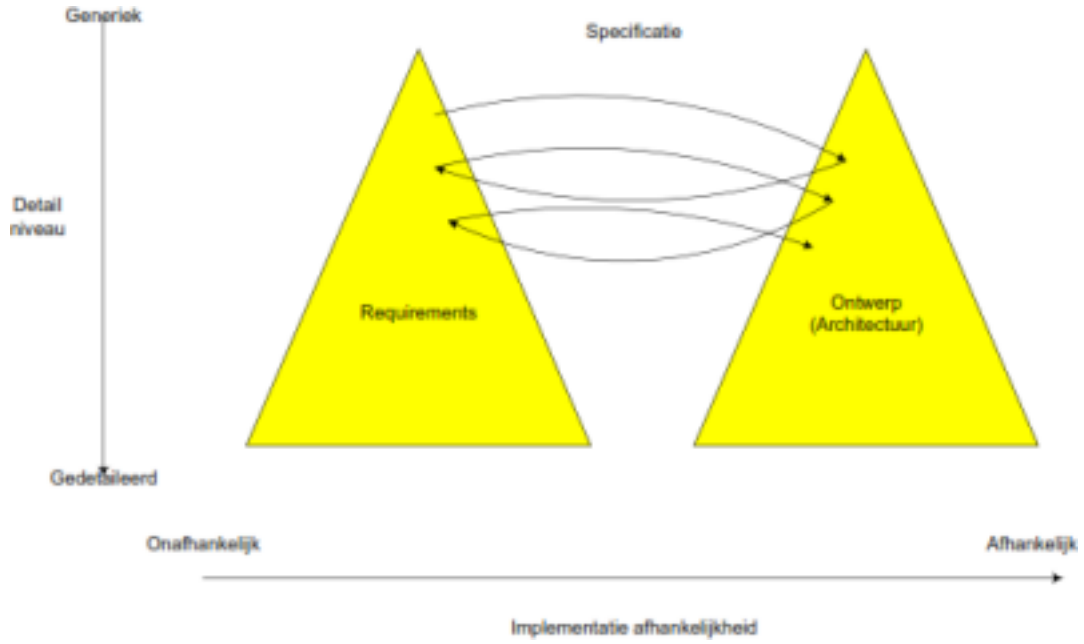
Bottom up : ***« pourquoi? »***

- Remettre en cause l'existant et les solutions préconisées
- Identifier les problèmes
- S'abstraire des détails
- Généraliser au niveau domaine

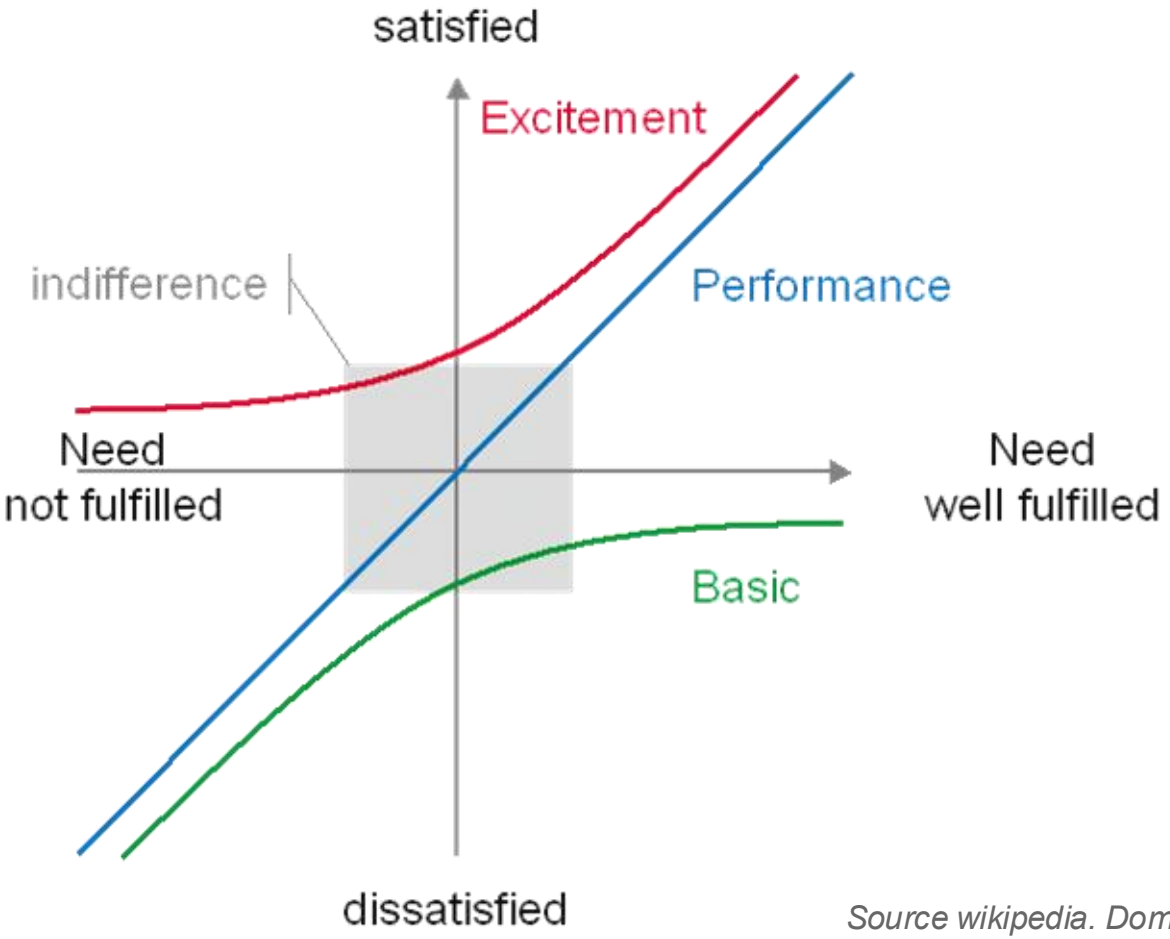
Top down : **« comment? »**

- Utiliser le raisonnement analogique
- Être créatif implique « inventer » les exigences au besoin
- Identifier les paramètres des solutions déjà identifiées et les décliner
- Recombiner

Twin Peak model



The **Kano model**



Source wikipedia. Domaine public
http://fr.wikipedia.org/wiki/Mod%C3%A8le_de_kano

Enquiry techniques

Goal :

- Elicit explicit knowledge

Means:

- Ask users their needs for the future system

Limitations :

- Motivation of stakeholder
- Ability to coomunicate
- Bias due to interviewer (ability to listen, transformation of message, suggestions, etc)

Practices:

- Individual or collective interviews
- Questionnaire based surveys



[Pohl-Rupp 2011] "Requirements Engineering Fundamentals - CPRE IREB"

Individual interviews VS surveys

- Why?
- How?
- What?
- When?
- Who?

…







Source : connectingdirectors.com

Creativity techniques

Goals :

- Develop innovative requirements

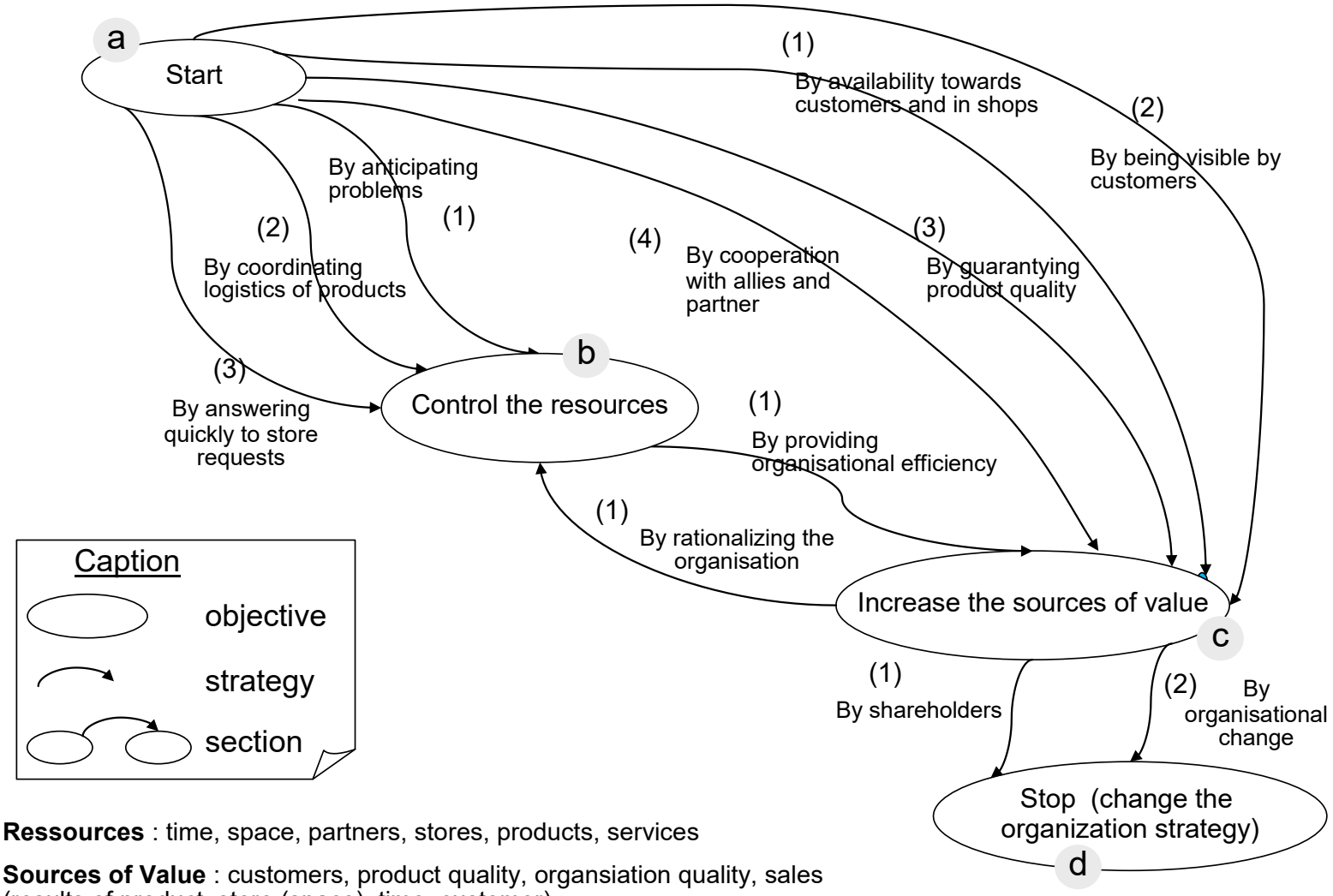
Pratices:

- **Brainstoming**
- **Brainstoming paradox**
- **Perspective modification**– « 6 thinking hats » [DeBono 2006]..
- **Analogies** (eg « metaphors » in « Extreme Programming »)



Source : [Pohl-Rupp 2011] "Requirements Engineering Fundamentals – CPRE IREB"

OR goal refinment



Ressources : time, space, partners, stores, products, services

Sources of Value : customers, product quality, organsiation quality, sales (results of product, store (space), time, customer)

Support techniques – **mind maps**

Goal :

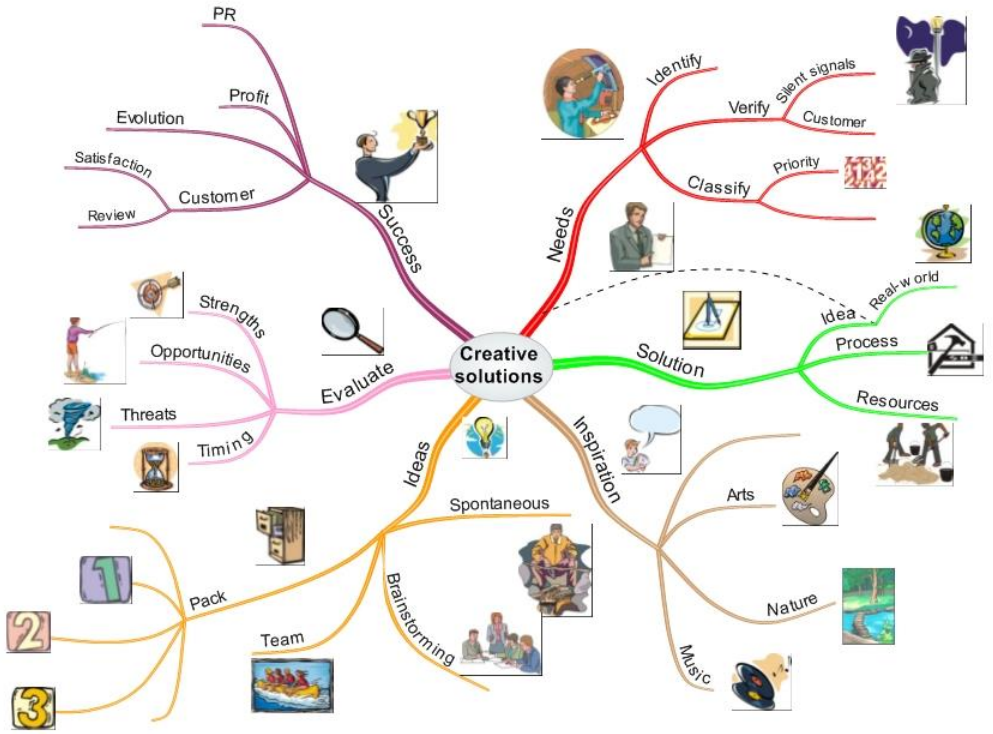
- Collect more ideas

Means:

- Keep track of flow

Practice:

- With a tool pr manually



[Pohl 2010]
« Requirements Engineering » §23.5

Workshops

Goal :

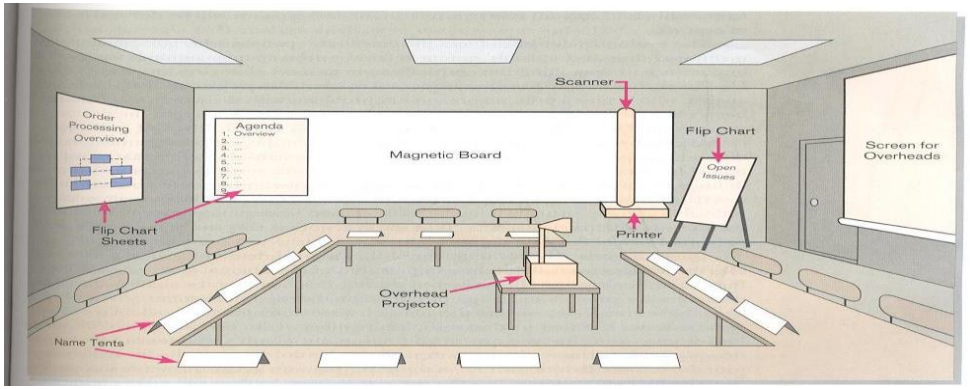
- Develop a shared vision and a group dynamics
- Elicit abstract knowledge
- Develop innovative requirements

Limitations :

- Time consuming

Practices:

- **focus** on a clear goal (eg using product chart)
- Establish clear working **rules** (facilitation)
- Prepare and communicate on the **agenda**
- Limit the **size** of the workshop (eg 5 people)
- Use a **parking** for good ideas
- **Time box** each step
- Keep highly **committed** people



Eg JAD workshop room

Requirements by collaboration – workshops for defining needs, Ellen Gottesdiener, [Gottesdiener 2002]

Legacy analysis

Goal :

- Understand the current situation
- Exploit requirements and knowledge that are already documented

Means:

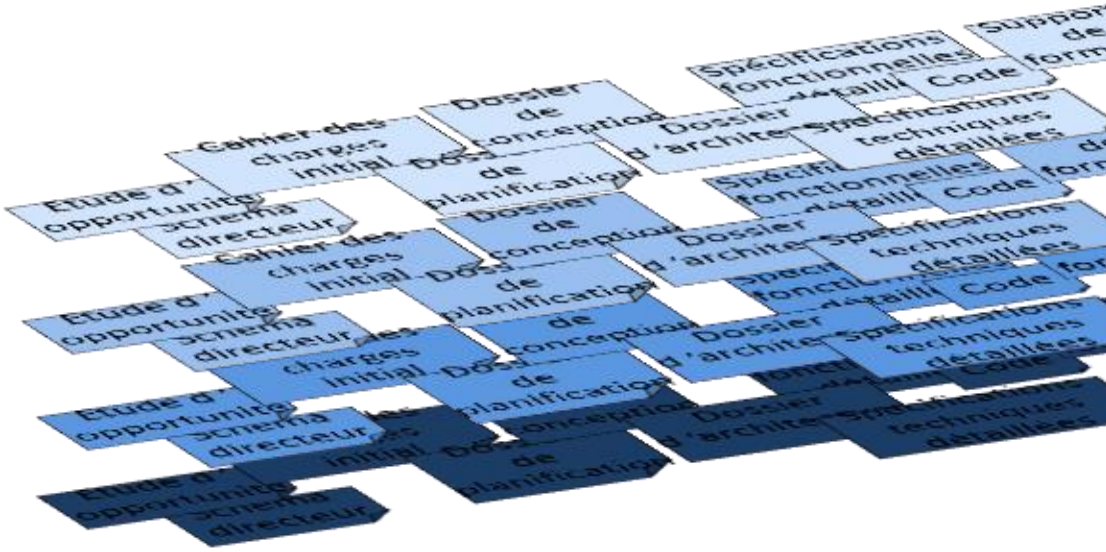
- Gather, evaluate, fitler information (plans, code, diagrams) or

Limitations :

- Needs critical view

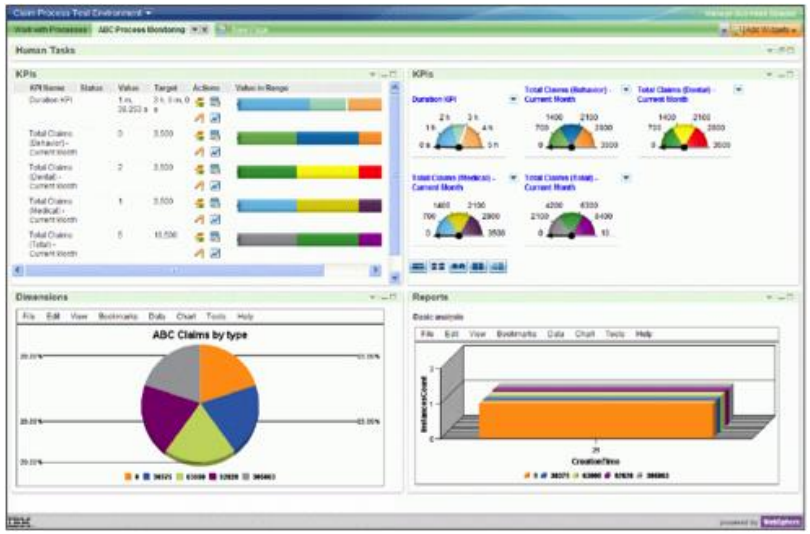
Practices:

- System archeology
- « perspective-based » reading
- Requirements reuse



Source : [Pohl-Rupp 2011] "Requirements Engineering Fundamentals – CPRE IREB"

Cost and performance analysis



- Process execution time and frequencies
- Productivity issues, idle time, bottelnecks
- Latency, size of work piles, communication delays
- Error rates
- etc

Source: www.ibis-instruments.com

User observation

Goal :

- Elicit tacit knowledge
- Understand the AsIs
- Adequate when users cannot express their needs

Means:

- Situational observation

Limitations :

- Not very efficient beyonf the As-Is
- Expensive and time consuming

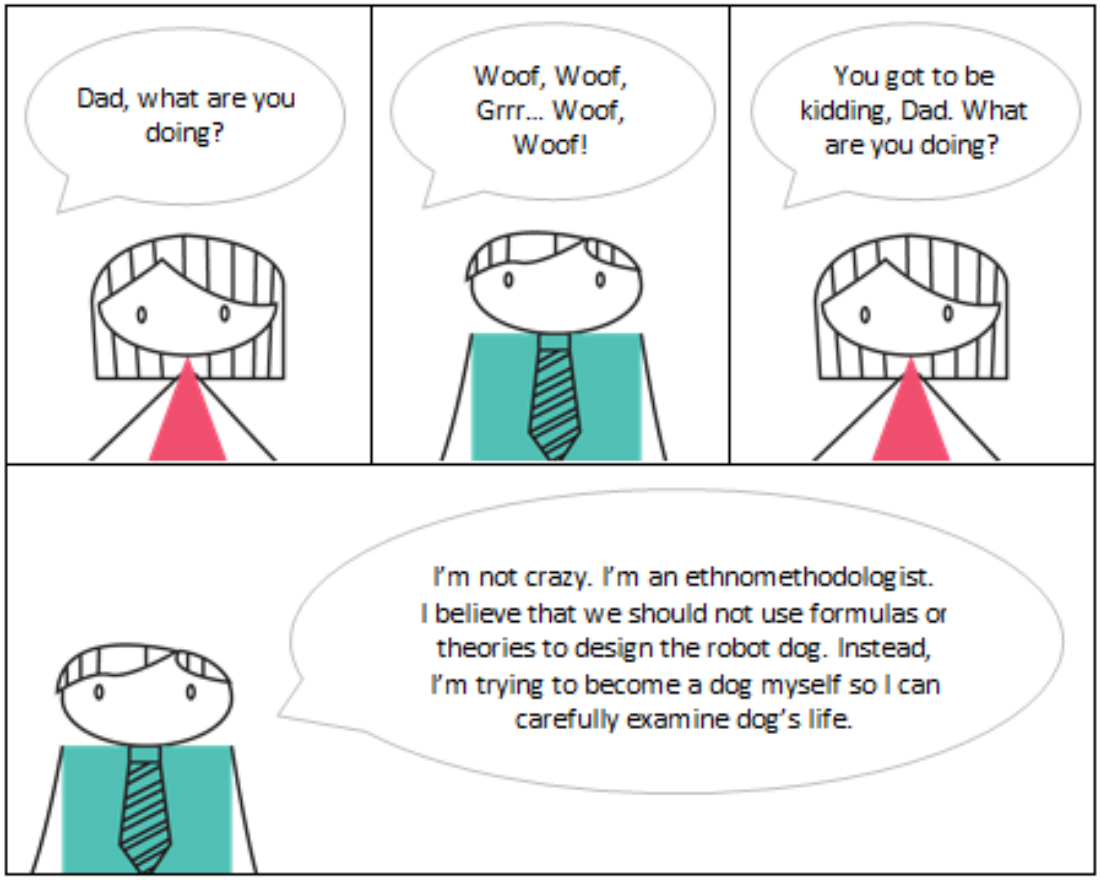
Practices:

- Field observation
- « quick and dirty » ethnography
- Conversation analysis



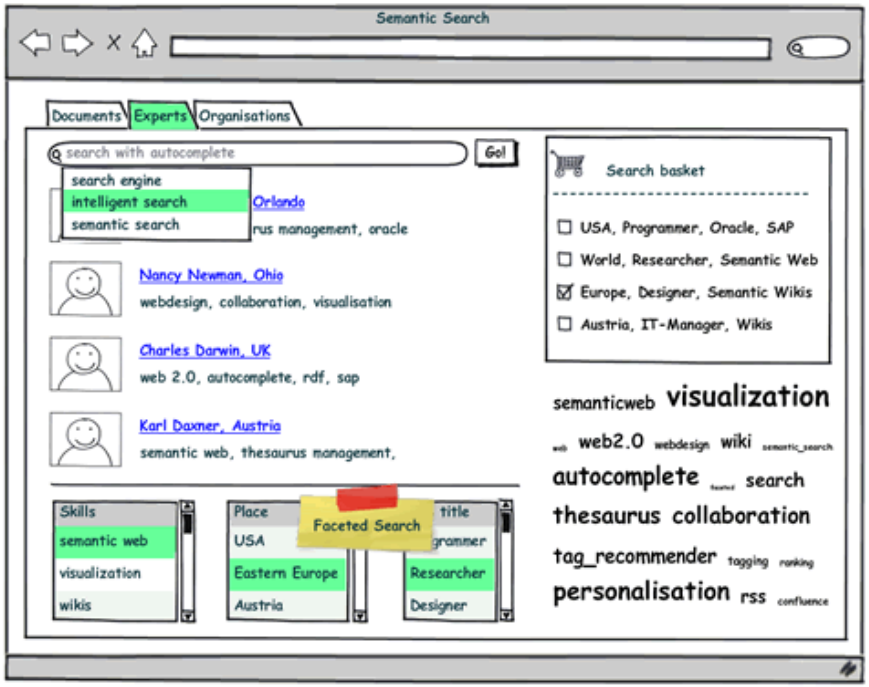
Source : [Pohl-Rupp 2011] "Requirements Engineering Fundamentals – CPRE IREB"

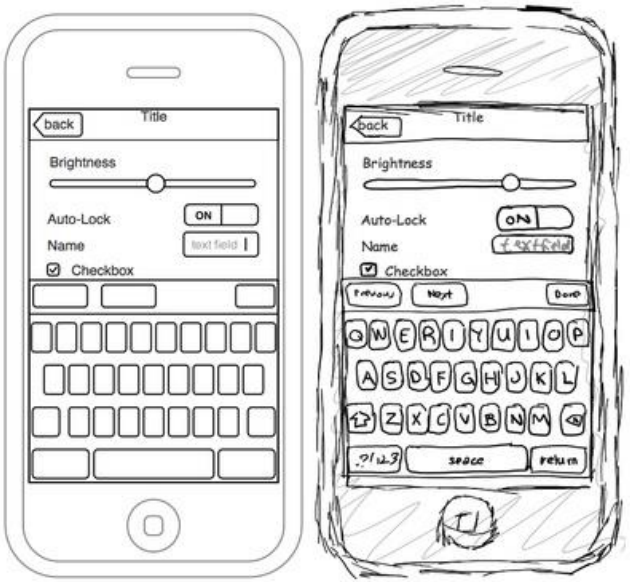
« quick and dirty » Ethnography

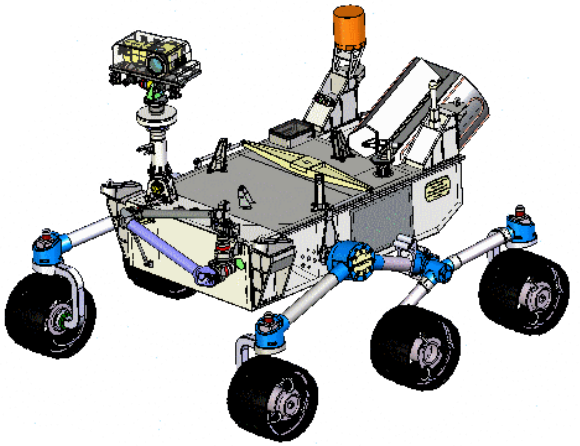


Source: theoryfamily.files.wordpress.com

Mock ups





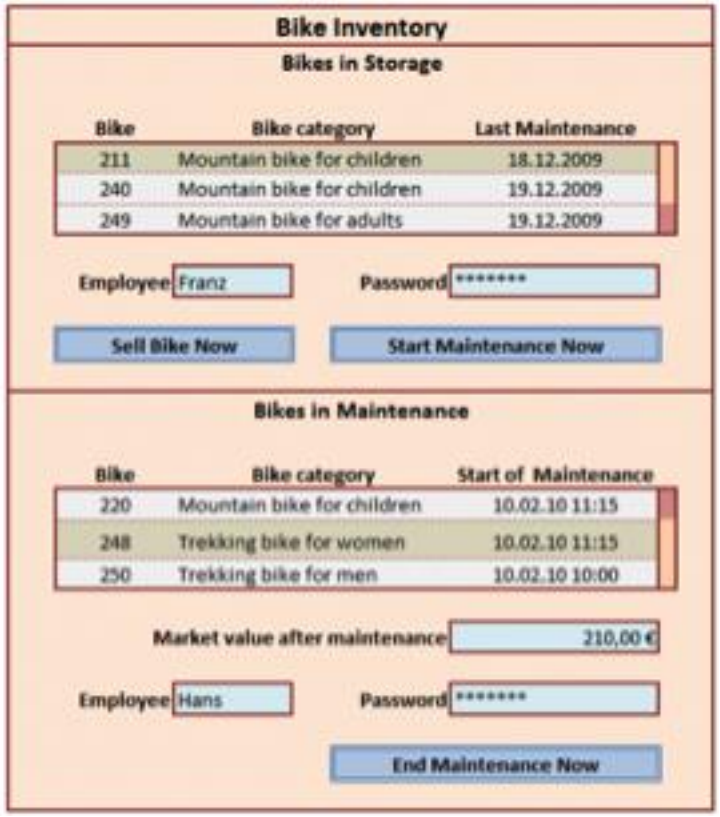


Soources : pedagogie.ac-toulouse.fr
iphoneized.com

Prototypes


Reserve Bikes
Username hans
Password *******
Login
Start 01.04.2010
End 03.04.2010
Bike category
Number
Mountain bike for children 1
Mountain bike for adults
Trekking bike for children
Trekking bike for women 2
Trekking bike for men 2
Reserve
Cancel

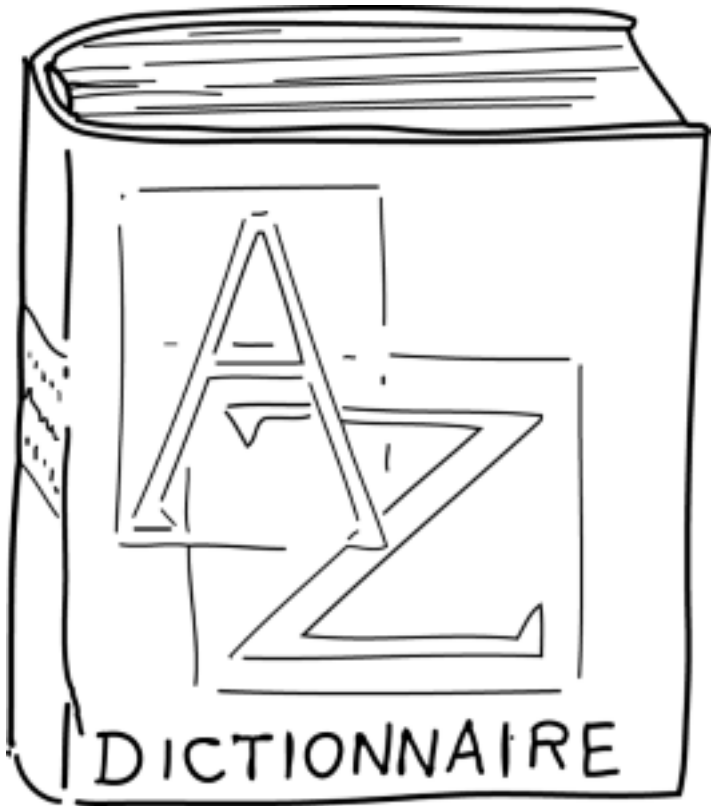

Your Reservations
Username hans
Password *******
Logout
Reservation
Start
End
Bikes
58447 01.04.2010 03.04.2010 5
58327 20.03.2010 20.03.2010 1
Cancel
Cancel


Bike Inventory
Bikes in Storage
Bike
Bike category
Last Maintenance
211 Mountain bike for children 18.12.2009
240 Mountain bike for children 19.12.2009
249 Mountain bike for adults 19.12.2009
Employee Franz
Password *******
Sell Bike Now
Start Maintenance Now
Bikes in Maintenance
Bike
Bike category
Start of Maintenance
220 Mountain bike for children 10.02.10 11:15
248 Trekking bike for women 10.02.10 11:15
250 Trekking bike for men 10.02.10 10:00
Market value after maintenance 210,00 €
Employee Hans
Password *******
End Maintenance Now

Other elicitation techniques

- Control lists for requirements elicitation (checklist)
  - Stakeholder list
  - Type of requirements list
- Modelling
  - Conceptual modelling
  - Simulation
- Reuse
  - Domain model
  - Ontonoly

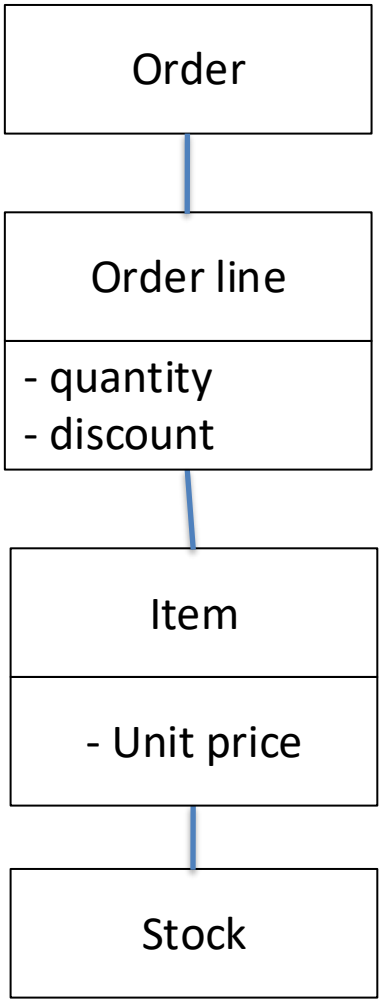
Domain dictionary



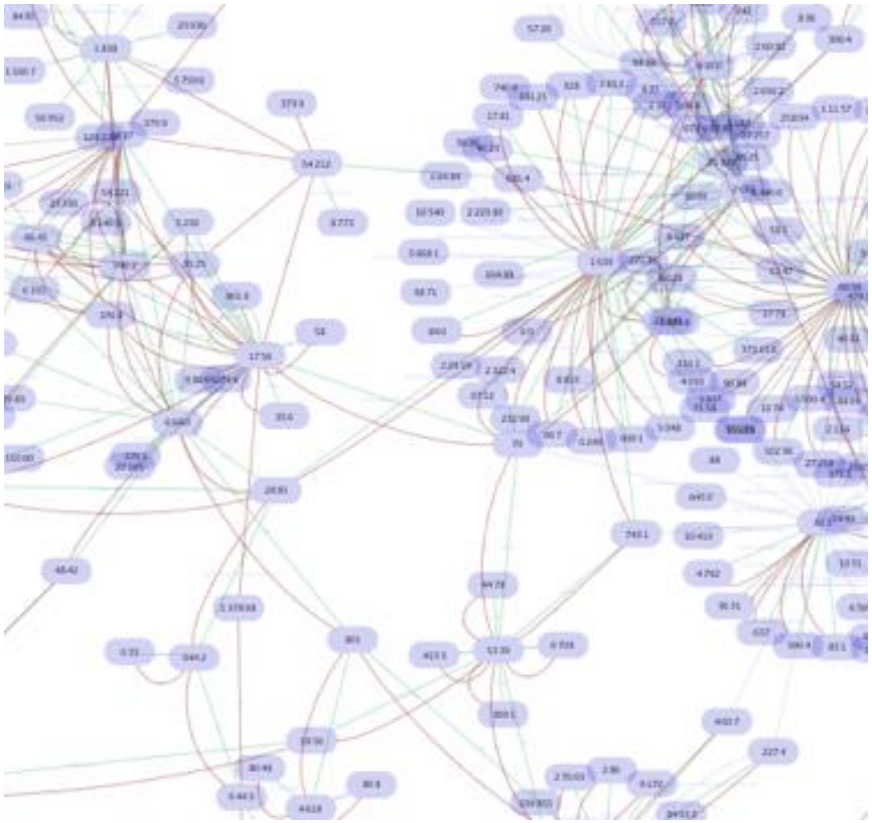
- Terms: actors, ressources, tasks information, rules, etc
- Acronyms, abreviations
- Definitions
- References/Sources
- Author

Domain models

- Formalized description of real world
- Shared by all
- Not implemented (abstract only)
- Uses a notation (E/R, BPMn, etc)

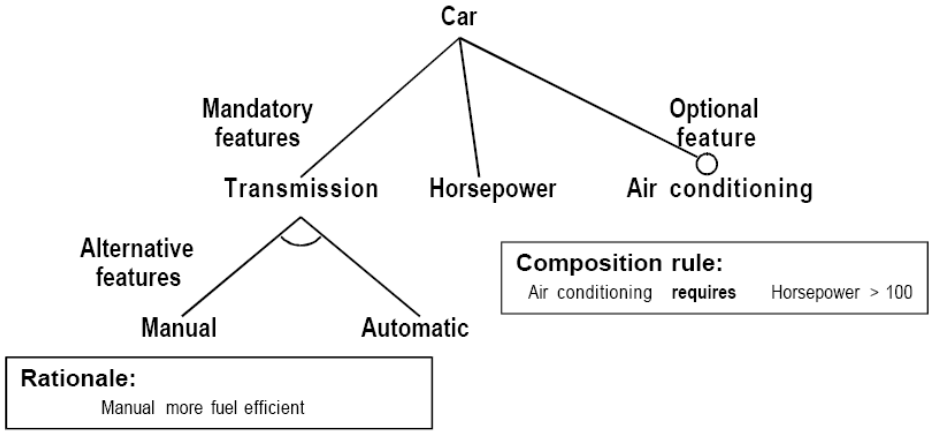


Ontologies



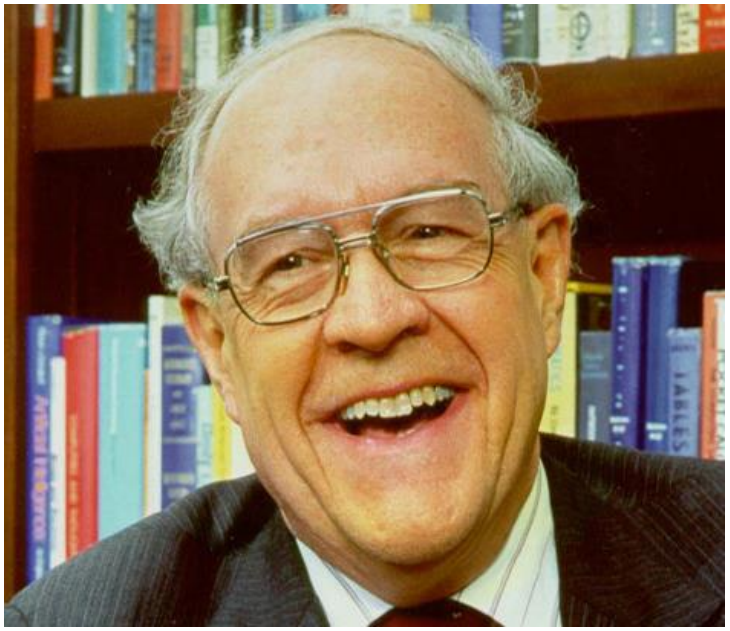
- Notations
  - Formal network of concepts
  - Logical rules
- Tools
  - Term extractors
  - Editors/modelers
  - Mapping tools
  - Solvers

Product lines



- Domain features
  - Optional
  - mandatory
- Dependencies
  - Sub-features
  - Exclues/requires
  - Individual and group cardinalities
- Links with reusable artefacts
  - Requirements
  - Models
  - Code
  - Test data, etc

Takeaway

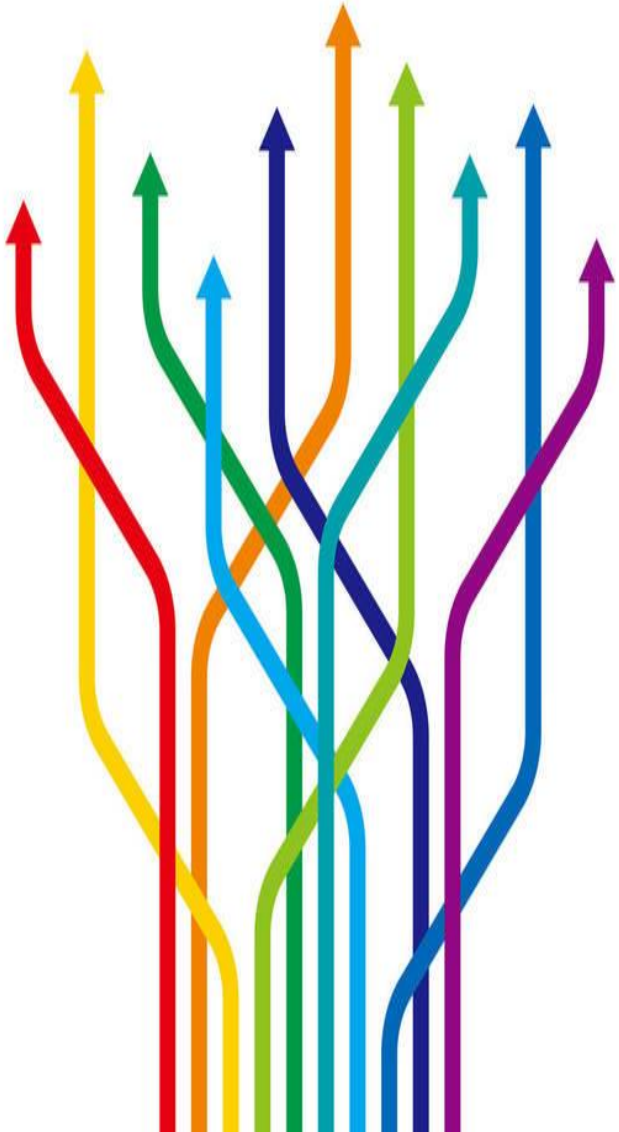


*"The **hardest** single part of building a software system is **deciding precisely what to build**"*

Fred Brooks

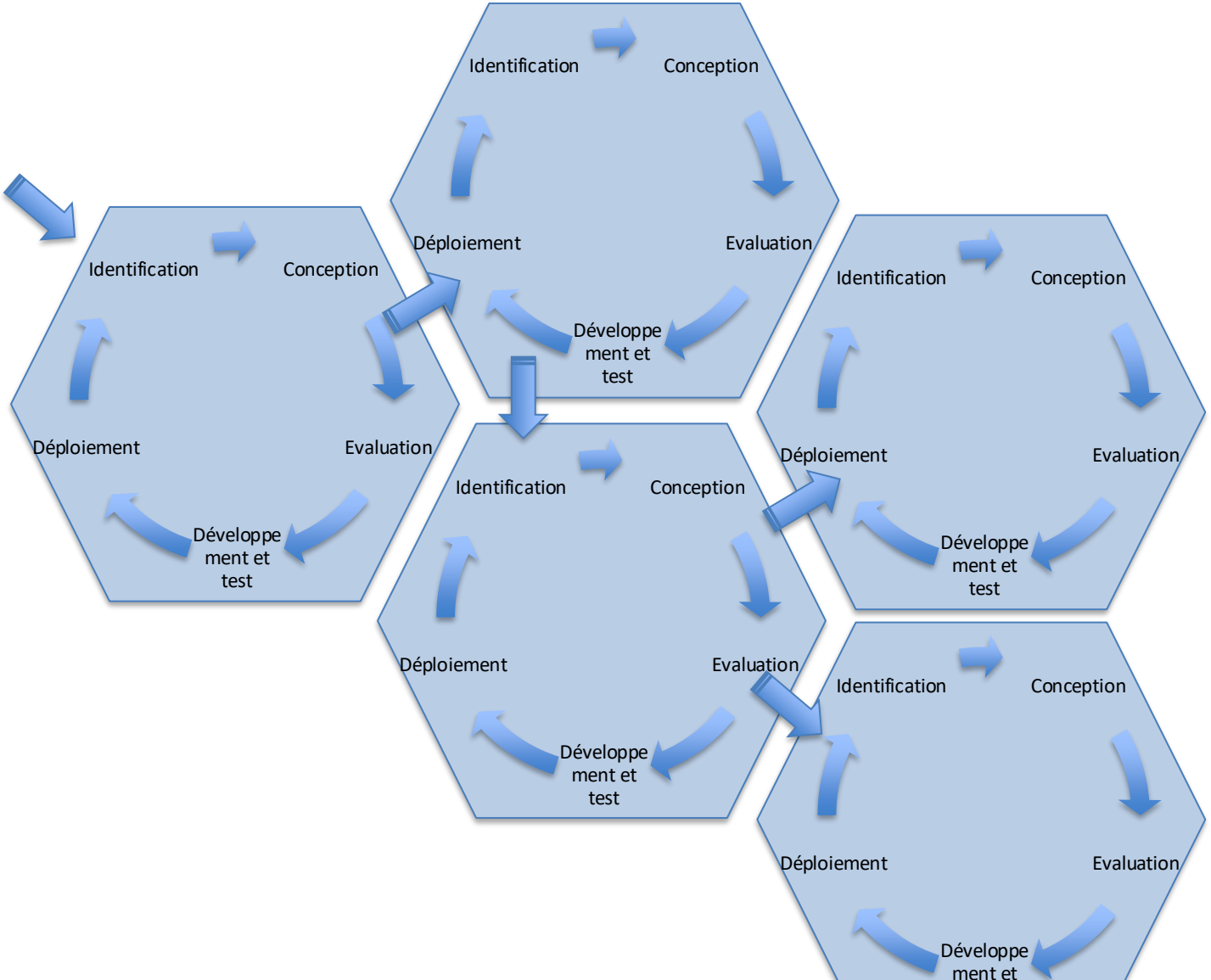
Prix Turing 2000

Factors influencing the choice of a requirements elicitation technique

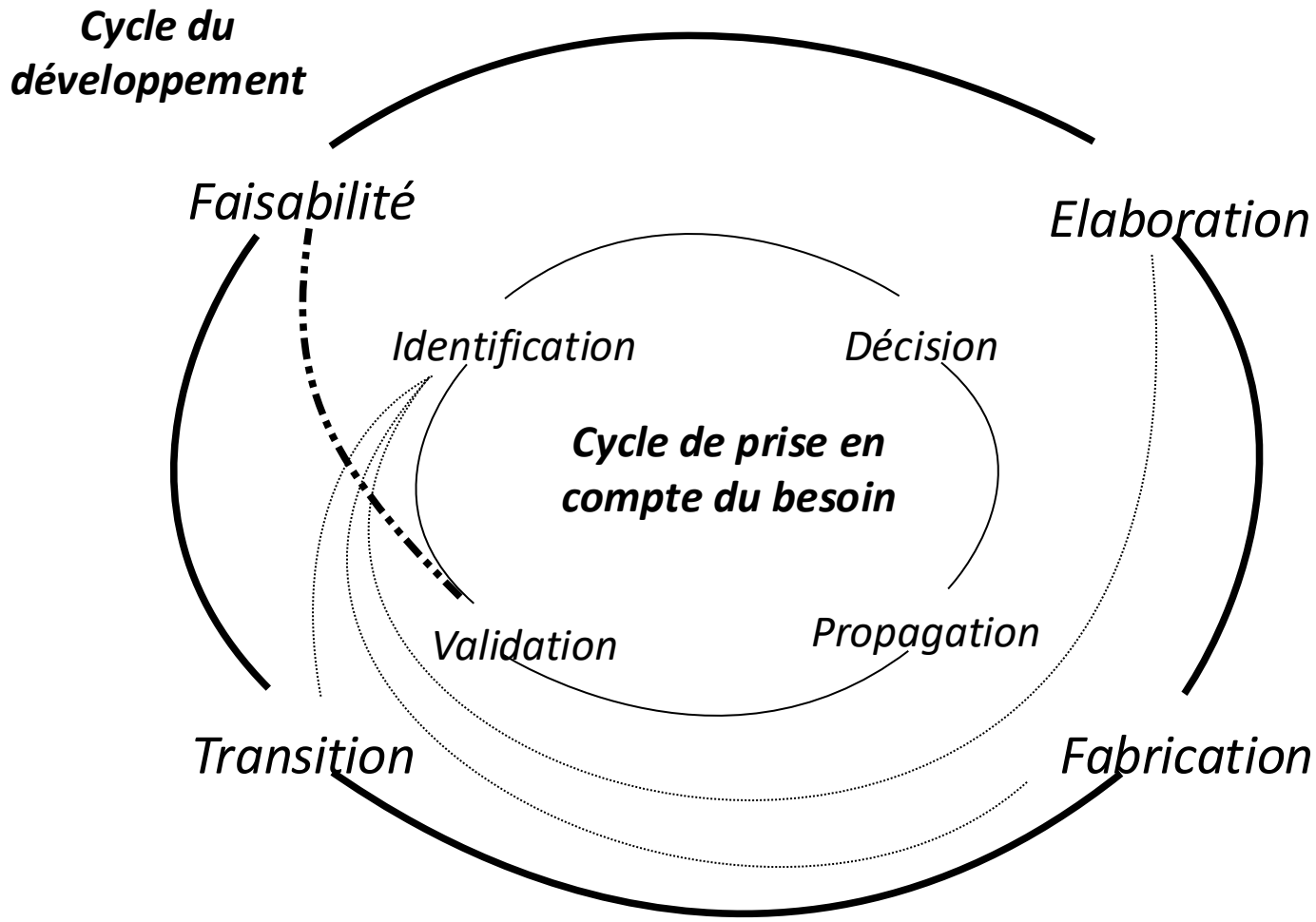


- Risk factors
- Human influences
- Organizational influences,
- Function-content influences
- Intended level of detail of the requirements

Be iteratie AND incremental



Don’t forget cycles interact



Terms

- none